ISCN- International System for Human Cytogenetic Nomenclature:

Total number (of centromeres),Sex chromosomes (comma, no space)

; to separate chromosomes or bands (inside brackets)
/ mosaicism
+ or -
del
dup
ins
inv
t
der
i

ISCN- International System for Human ytogenetic Nomenclature:

Total number (of centromeres),Sex chromosomes (comma, no space)

69,XXX
46,XX
47,XY,+21
47,XY,+21/46,XY
45,XY,-13,-14,t(13q;14q)
46,XX,t(4;12)(q12;p13)
46,XY,dup(5p)
46,XX,del(4)(p14;pter)
46,XY,del(5p)
46,X,i(Xq)

https://learn.genetics.utah.edu/content/basics/karyotype/

https://biology.arizona.edu/human_bio/activities/karyotyping/karyotyping.html

Types of Aneuploidy

- **Trisomy:** Having an extra copy of a chromosome (totaling 47):
  - *Trisomy 21 (Down syndrome):* Extra chromosome 21
  - *Trisomy 18 (Edwards syndrome):* Extra chromosome 18
  - *Trisomy 13 (Patau syndrome):* Extra chromosome 13

- **Monosomy:** Missing a copy of a chromosome (totaling 45):
  - *Turner syndrome (45, X):* Missing or incomplete X chromosome in females.

- **Sex Chromosome Aneuploidies:** Abnormalities involving the X or Y sex chromosomes:
  - *Klinefelter syndrome (47, XXY):* Males with an extra X chromosome.
  - *Triple X syndrome (47, XXX):* Females with an extra X chromosome.

Prevalence of chromosomal aberrations in spontaneous abortions and stillborns:

Until week 7:	20%
Weeks 7 to 20:	35%
Weeks 20 to 40:	4%

Prevalence of aneuploidies (per 1000 live newborns):

47,XX,+21	1.5
47,XX,+18	0.12
47,XX,+13	0.07
45,X	0.4
47,XXY	1.5
47,XYY	1.5

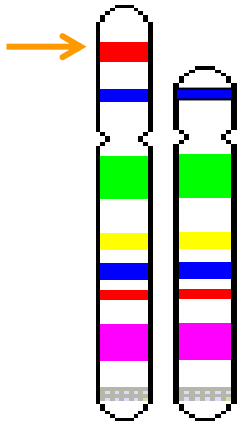


Maternal age	DS risk
15	1:1578
20	1:1528
25	1:1351
30	1:909
35	1:384
40	1:112

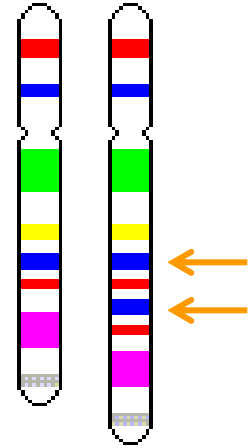
47,XX,+21

46,XX/47,XX,+21

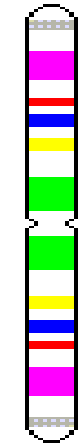
TERMINAL DELETION
del(7)(p22;pter)

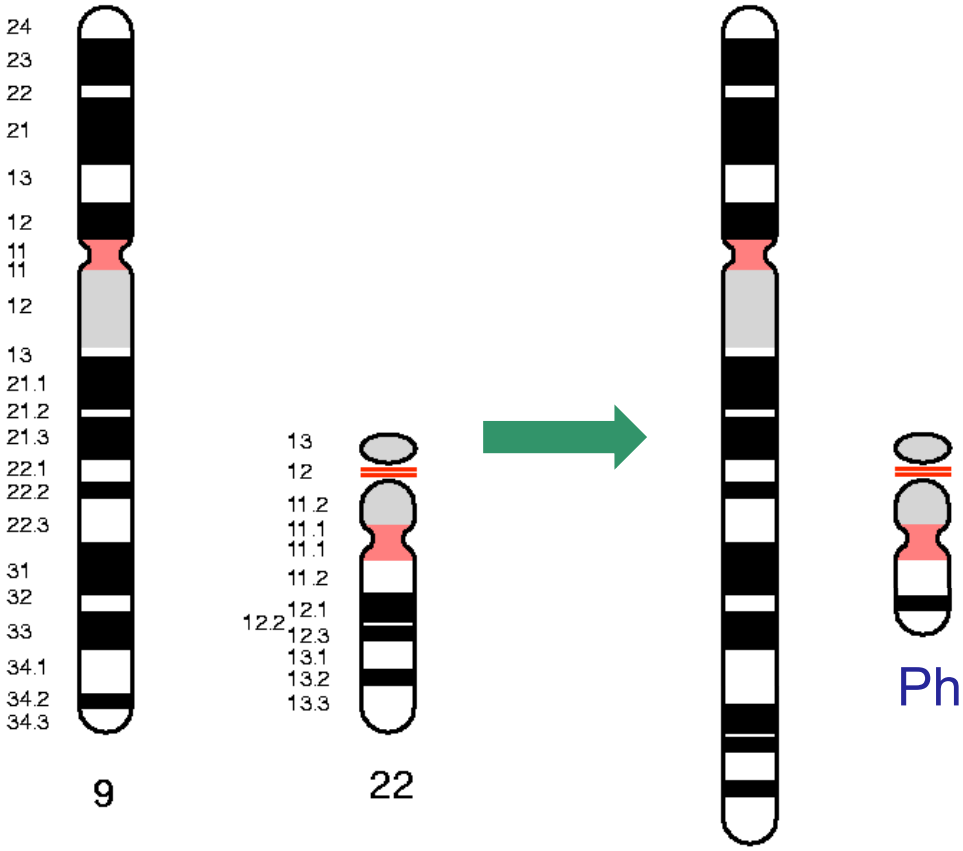


DUPLICATION
dup(7)(q21;q23)



ISOCHROMOSOME
46,X,i(Xq)



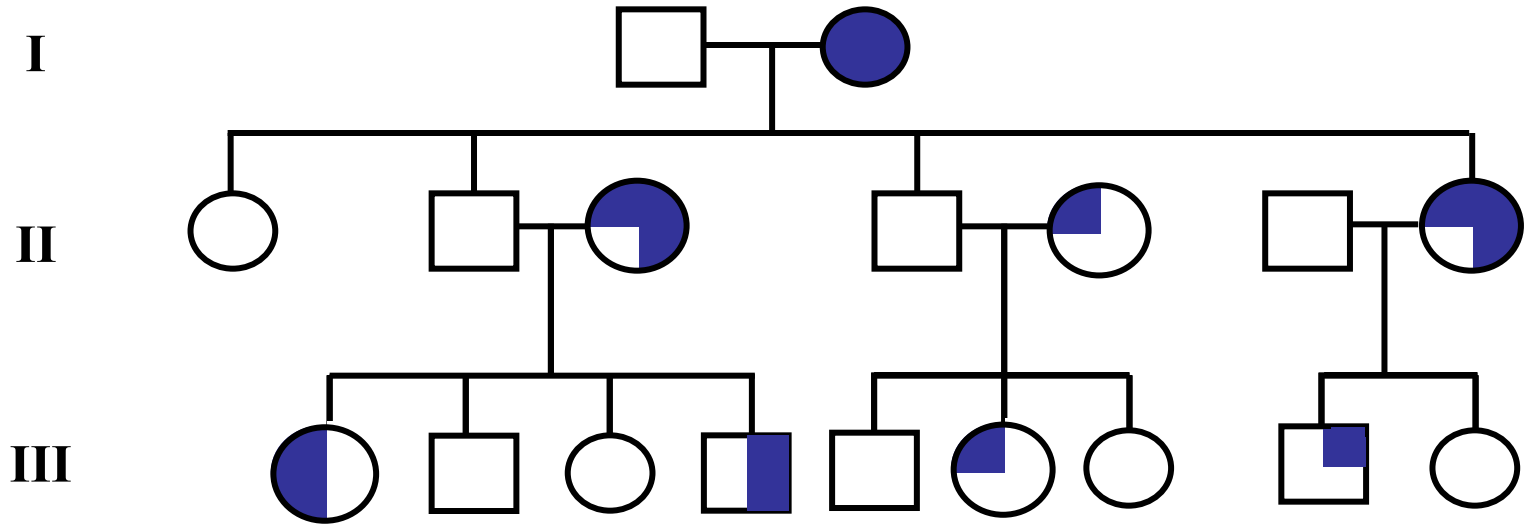


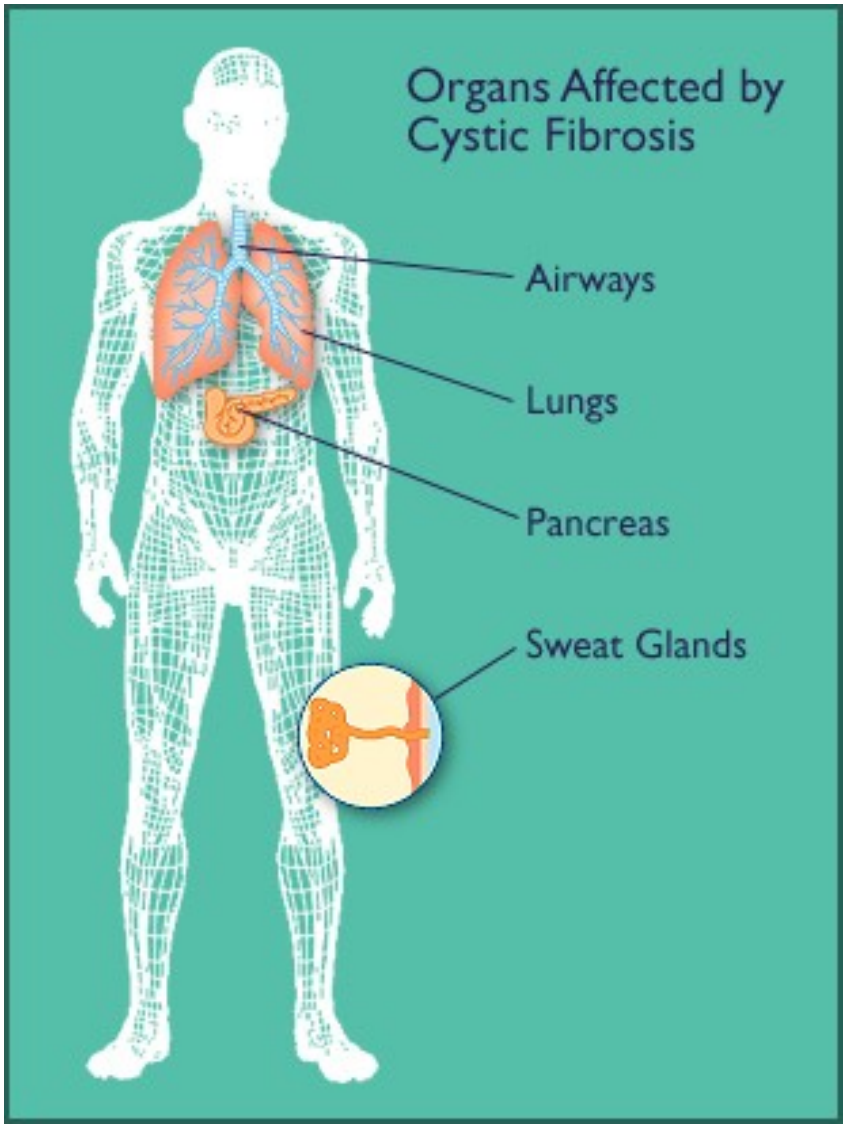
t(9;22)(q34;q11)

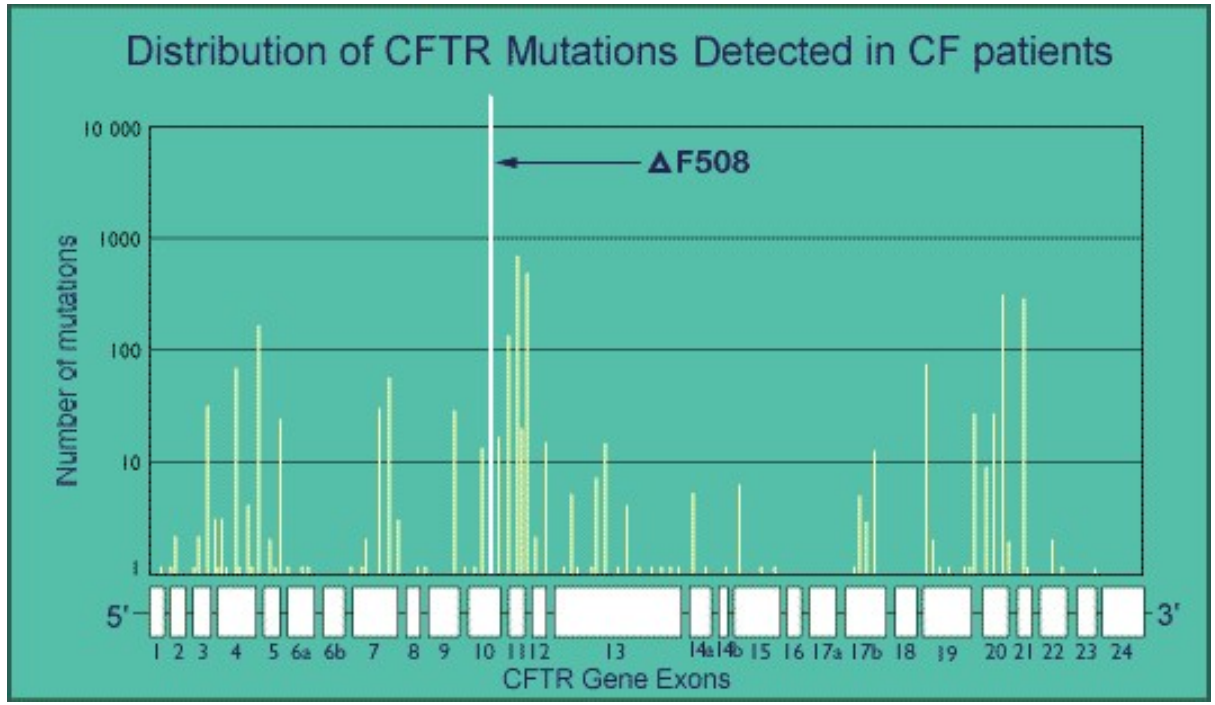
Chronic Myeloid Leukemia

Philadelphia chromosome (der 22)

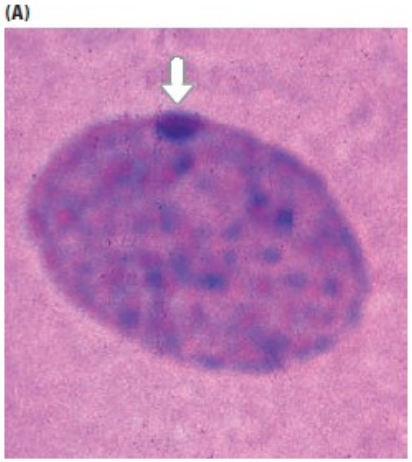
Variable expressivity

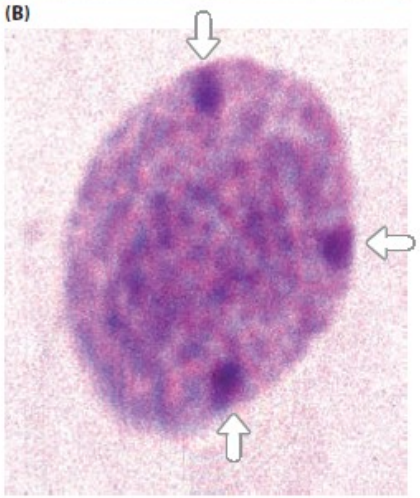


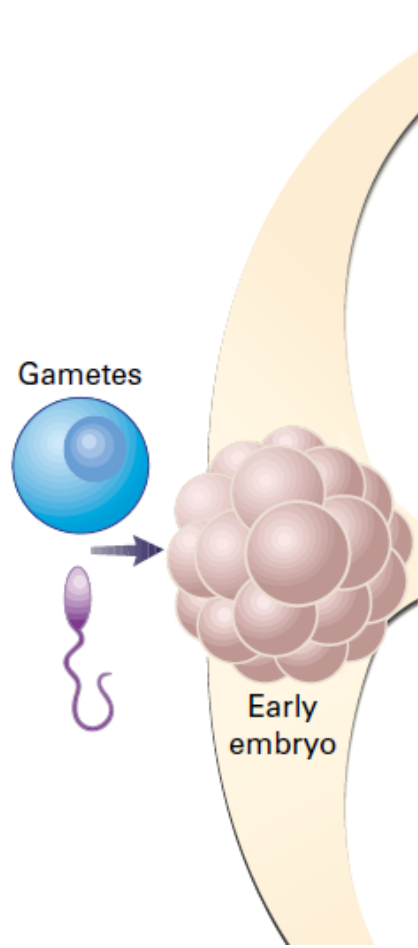

Organs Affected by Cystic Fibrosis
Airways
Lungs
Pancreas
Sweat Glands

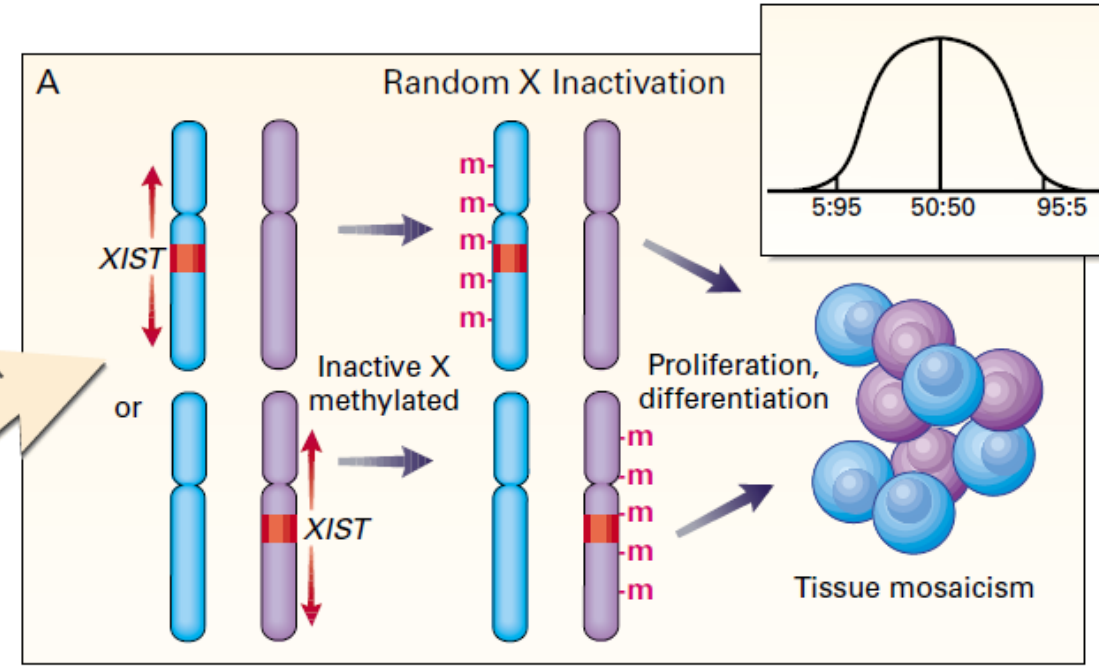

Distribution of CFTR Mutations Detected in CF patients
ΔF508
Number of mutations
10 000
1000
100
10
1
5'
3'
1 2 3 4 5 6a 6b 7 8 9 10 11 12 13 14a 14b 15 16 17a 17b 18 19 20 21 22 23 24
CFTR Gene Exons

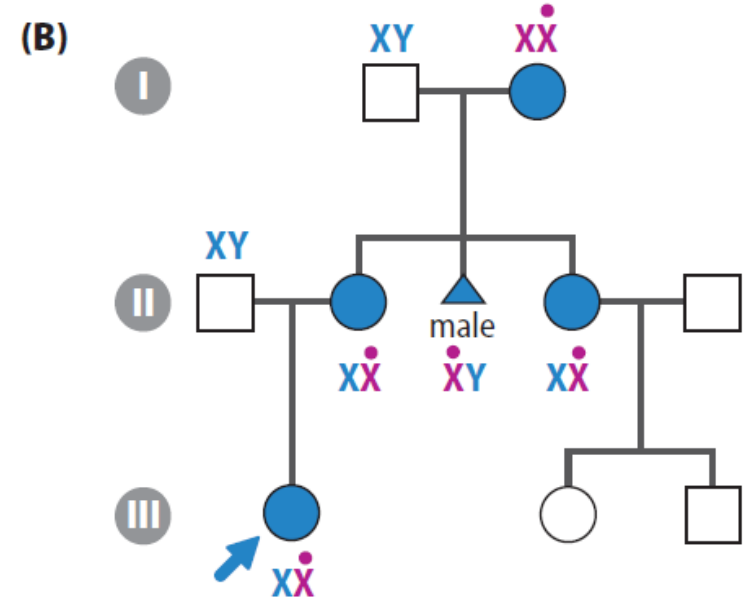
X-chromosome inactivation



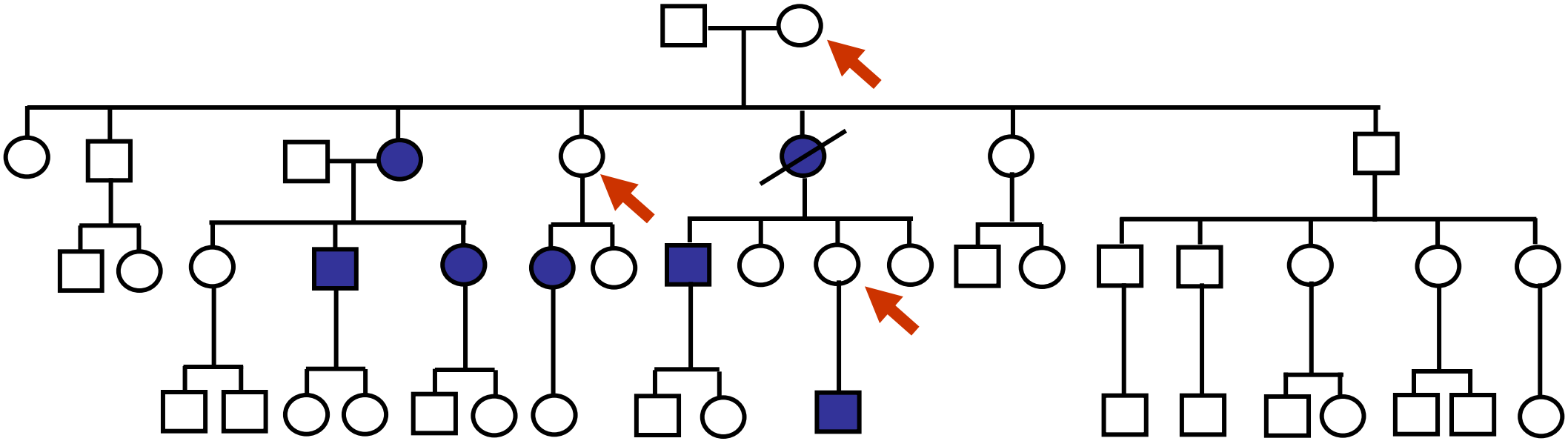








Mitochondrial inheritance



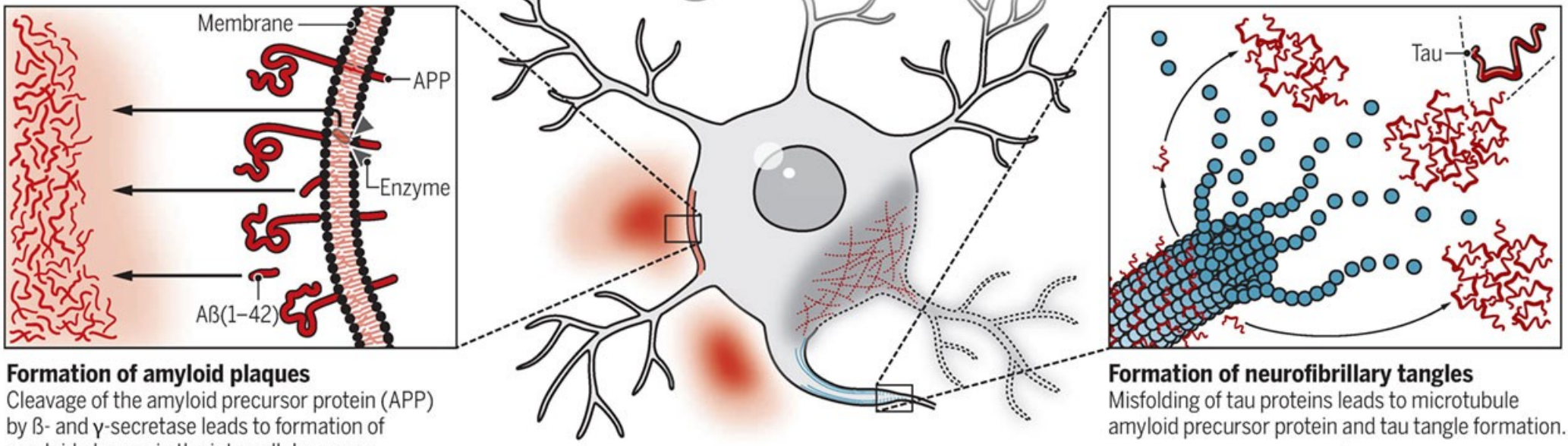
Males NEVER pass the disease

Females can pass the disease to sons and daughters

Some women are unaffected carriers

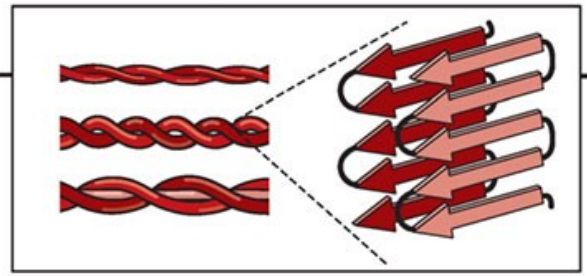
Molecular characteristics of Alzheimer's disease

Amyloid plaques and neurofibrillary tangles that accumulate in an Alzheimer's brain consist of amyloid fibrils with different components but similar tertiary protein structures.



Formation of amyloid plaques
Cleavage of the amyloid precursor protein (APP) by ß- and γ-secretase leads to formation of amyloid plaques in the intercellular space.

Formation of neurofibrillary tangles
Misfolding of tau proteins leads to microtubule amyloid precursor protein and tau tangle formation.



Amyloid fibrils
Aß(1–42) and tau build various fibril structures that differ in number, orientation, and structure of strands.

Cross-ß sheet structure, amyloid state
Both tau and Aß(1–42) have a similar overall tertiary protein structure.

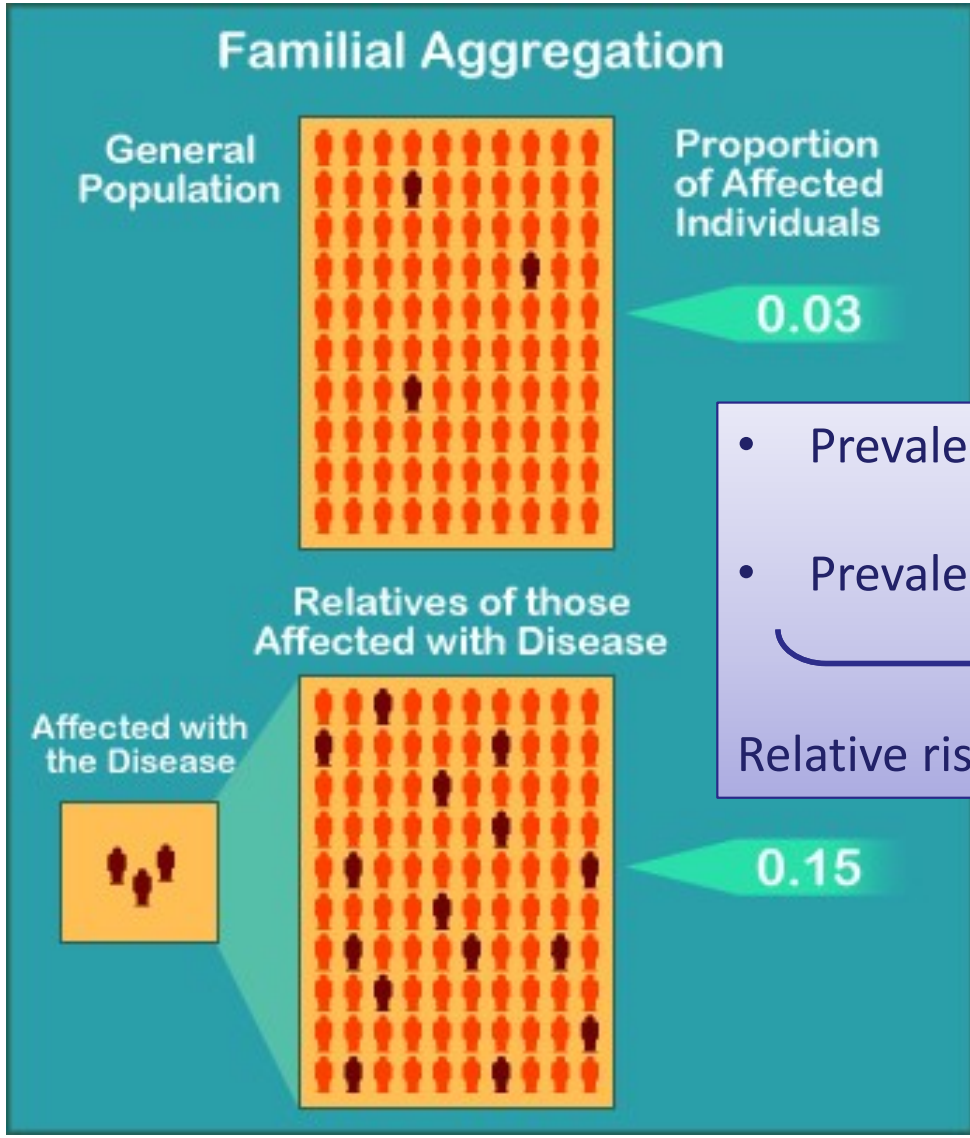


Table 2 | ***APOE* genotype, AD risk and amyloid-β deposition**

APOE genotype	Frequency in cognitively healthy individuals (%)[286]	Frequency in patients with AD (%)[286]	Odds ratio for AD development[9]
*APOE*ε2/ε2*	0.7	0.3	0.56
*APOE*ε2/ε3*	11.0	4.6	0.56
*APOE*ε3/ε3*	62.3	34.3	1.00
*APOE*ε2/ε4*	1.9	2.6	2.64
*APOE*ε3/ε4*	22.2	43.4	3.63
*APOE*ε4/ε4*	1.9	14.8	14.49

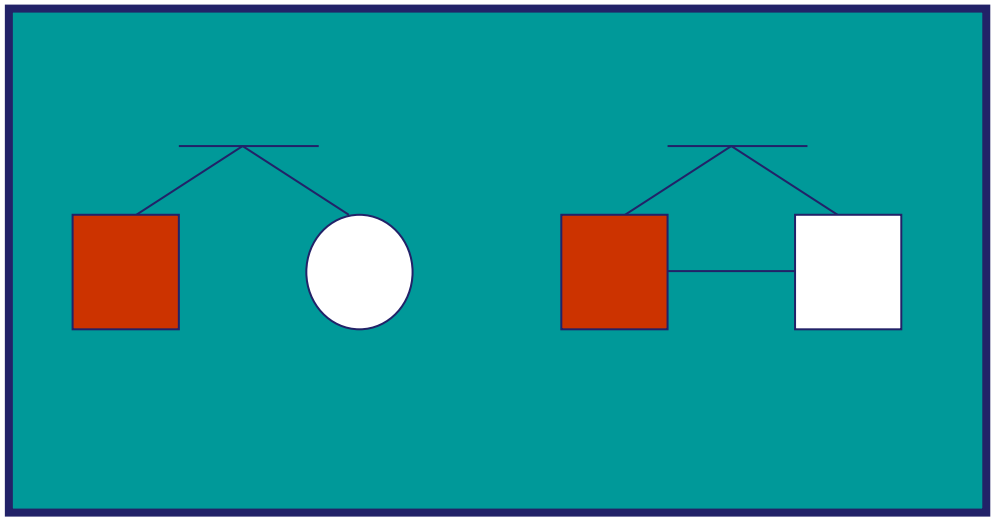
Alzheimer disease (AD)

- **Multiple pathogenicity**: Beta-amiloid (early onset <65 y.) and downstream pathways (late-onset >65 y., 95% of all patients).
- **Protective and risk variants**: the apoE4 model. The apolipoprotein E (APOE) gene has three major allelic variants, APOE*ε2, APOE*ε3 and APOE*ε4; APOE*ε4 is associated with an increased risk and lower age of onset of late-onset AD, whereas APOE*ε2 seems to confer protection against AD.
- Carrying **one** APOE*ε4 allele increases the risk of late-onset AD 3–4-fold, and carrying **two** alleles increases the risk 9–15-fold. ε2 decreases the risk to 0.6-fold (compared to ε3).



- Prevalence in general population: 3/100
- Prevalence in siblings (sibs): 15/100

Relative risk (sibs vs. general): $\lambda s = 15/3 = 5$



$$h^2 = 2 \times (C_{MZ} - C_{DZ})$$

Concordance Rates

Disease	MZ Twins	DZ Twins	Heritability
Measles			